An office building prepared for a digital twin

Three floors. Two office wings. A retained central atrium. Services planned from the start.

AURA / THREE-FLOOR OFFICE CONCEPT

Furnished cutaways | independent conceptual model based on the supplied screenshots

GROUND
Arrival + services



FIRST
Teams + collaboration



SECOND
CEO + boardroom



EXIT-SIGN & EGRESS OVERLAY / REV E01 / 28 OCT 2026

Plans on pages 2-4 now show proposed exit-sign positions and candidate routes. Read the design basis on page 7 and the sign schedules on pages 8-9 together with the plans.

User confirms NFPA 101 and existing enclosed rear Stairs A and B. The base plan still shows reservations. Stair geometry, exterior discharge, atrium protection and egress compliance remain to be verified.

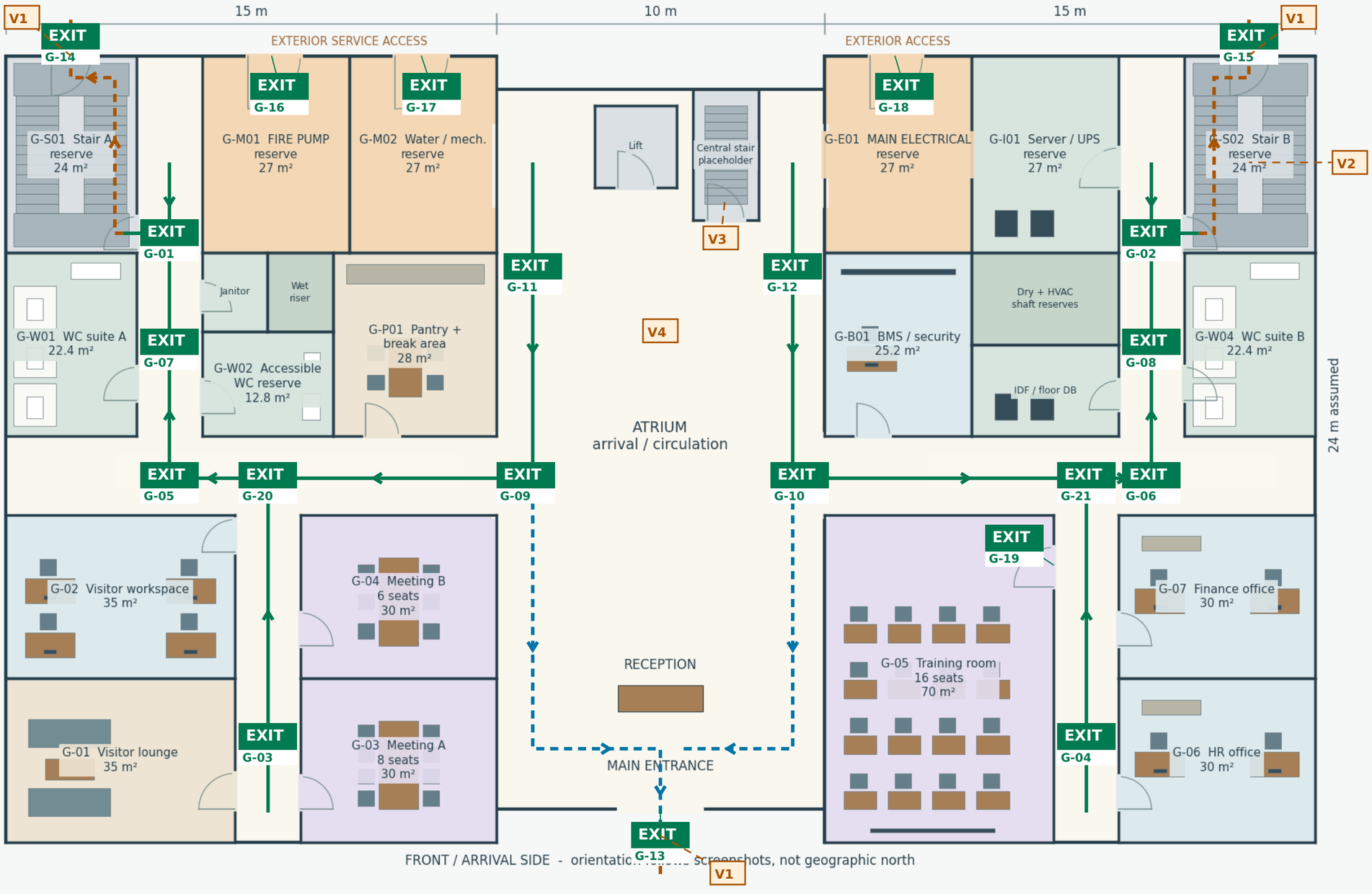
Ground: reception, visitor lounge, two meeting rooms, training, HR, finance and BMS/security.

First: team workspaces, project offices, meeting rooms and IT support. Second: CEO suite, boardroom and executive offices.

This is a new concept reconstructed from screenshots, not an edit of the supplied model. The 40 x 24 m envelope is an explicit working assumption; actual dimensions and column locations have not been measured.

Ground floor | Arrival + public functions

EGRESS CONCEPT / G | PROPOSED SIGN LOCATIONS | NFPA 101 basis; adopted edition and authority review pending



READ THIS OVERLAY EXIT ID Proposed illuminated EXIT sign; ID links to schedule (pp. 8-9). Travel toward rear stair Atrium-related path

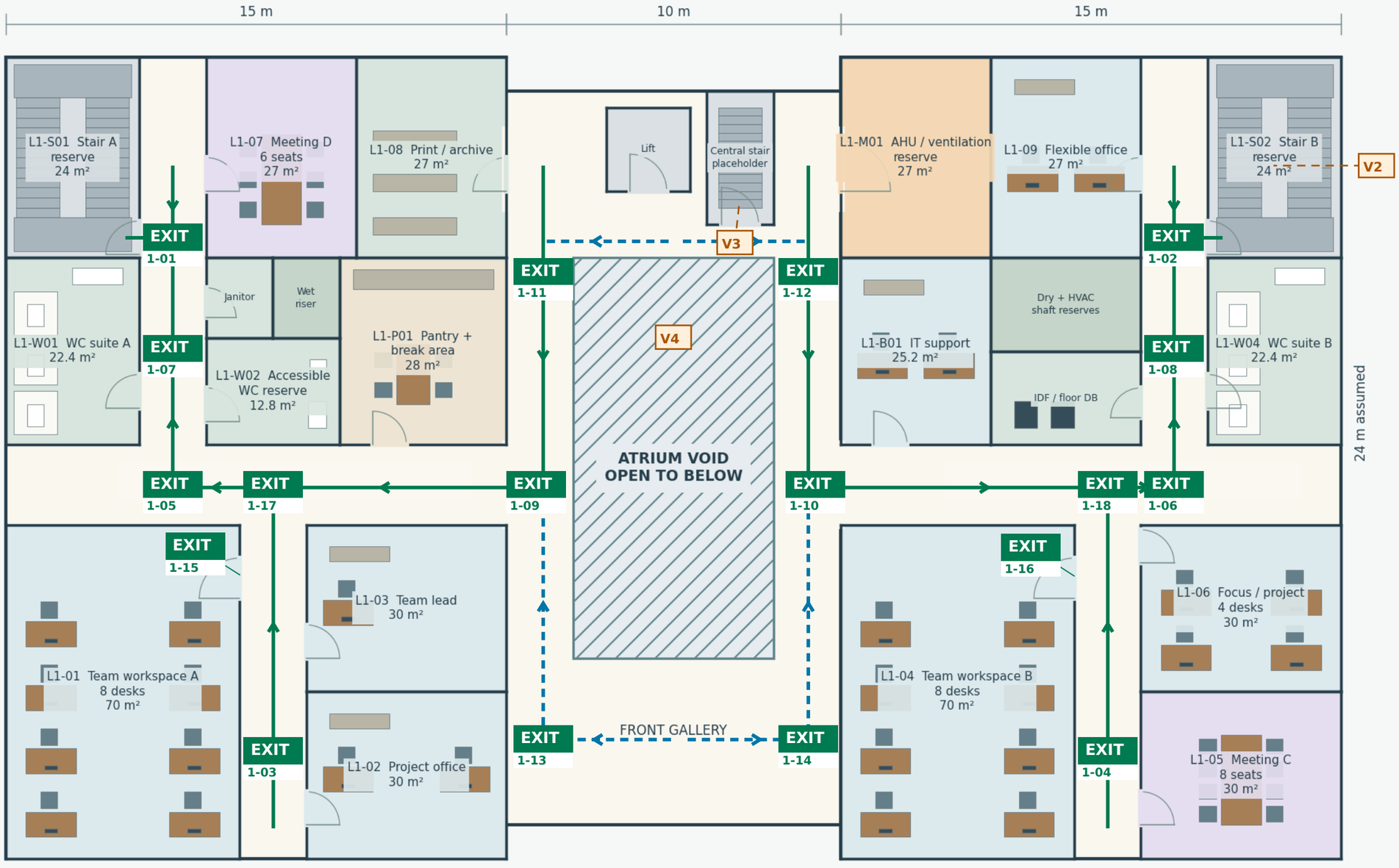
Arrows show travel on the PLAN, not the printed arrow on the sign. Final sign faces and chevrons must match the approach.

V1: stair discharge links / outdoor paths unverified. V2: enclosed stairs user-confirmed; survey doors, enclosure and capacity. V3: central stair + lift excluded.

V4: atrium fire/smoke strategy must validate all exposed routes. Room-door signs are selective; check sightlines from every occupied area.

First floor | Team offices + collaboration

EGRESS CONCEPT / 1 | PROPOSED SIGN LOCATIONS | NFPA 101 basis; adopted edition and authority review pending



FRONT / ARRIVAL SIDE - orientation follows screenshots, not geographic north

READ THIS OVERLAY EXIT ID Proposed illuminated EXIT sign; ID links to schedule (pp. 8-9). Travel toward rear stair Atrium-related path

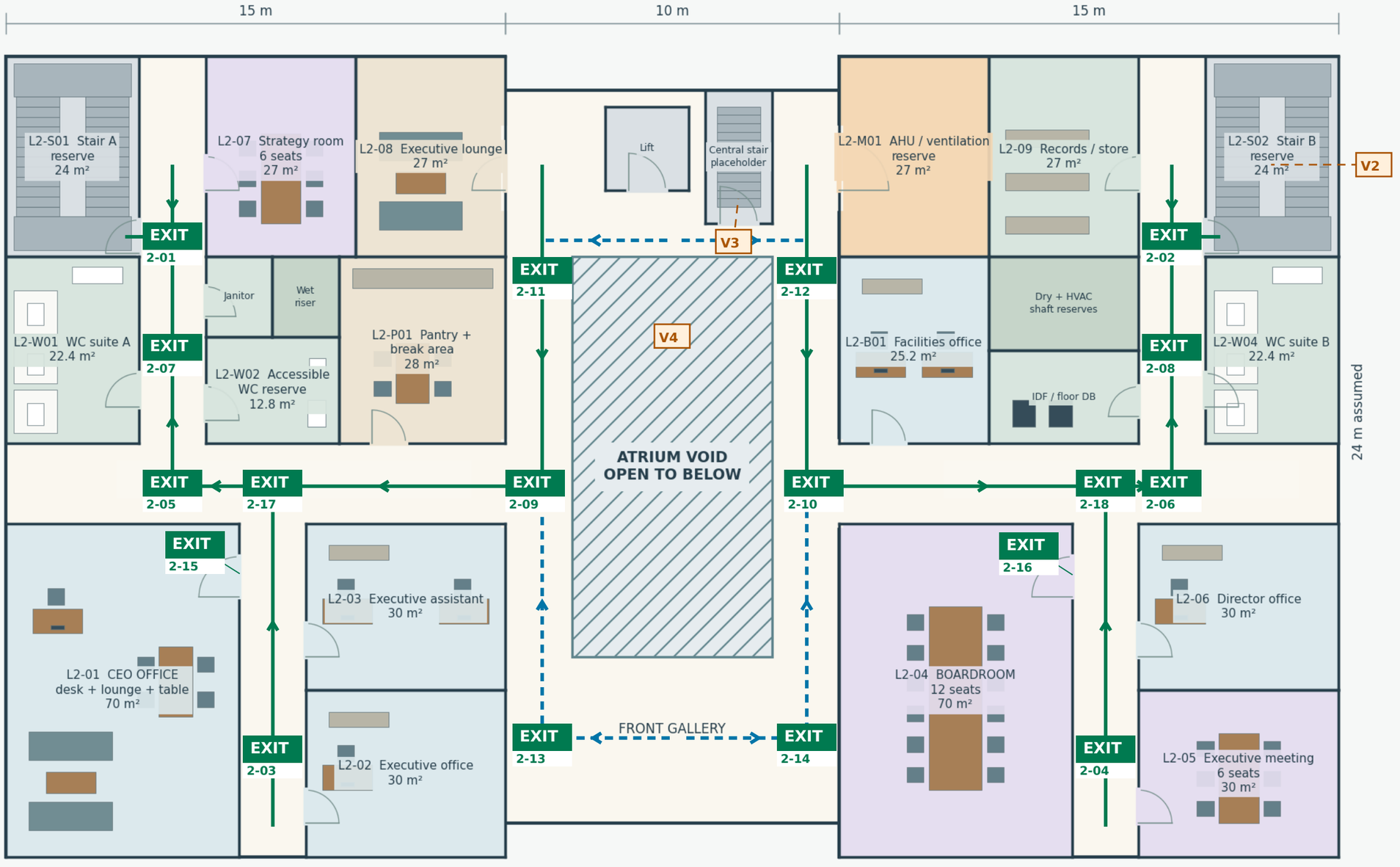
Arrows show travel on the PLAN, not the printed arrow on the sign. Final sign faces and chevrons must match the approach.

V1: stair discharge links / outdoor paths unverified. V2: enclosed stairs user-confirmed; survey doors, enclosure and capacity. V3: central stair + lift excluded.

V4: atrium fire/smoke strategy must validate all exposed routes. Room-door signs are selective; check sightlines from every occupied area.

Second floor | Leadership + boardroom

EGRESS CONCEPT / 2 | PROPOSED SIGN LOCATIONS | NFPA 101 basis; adopted edition and authority review pending



FRONT / ARRIVAL SIDE - orientation follows screenshots, not geographic north

READ THIS OVERLAY EXIT ID Proposed illuminated EXIT sign; ID links to schedule (pp. 8-9). Travel toward rear stair Atrium-related path

Arrows show travel on the PLAN, not the printed arrow on the sign. Final sign faces and chevrons must match the approach.

V1: stair discharge links / outdoor paths unverified. V2: enclosed stairs user-confirmed; survey doors, enclosure and capacity. V3: central stair + lift excluded.

V4: atrium fire/smoke strategy must validate all exposed routes. Room-door signs are selective; check sightlines from every occupied area.

Keep these spaces available

Reservations for the next modelling stage - neither equipment sizing nor an installation design.

SPACE	LOCATION	ALLOWANCE	DESIGN INTENT
Fire pump room	Ground, rear left	27 m2	Dedicated exterior door; no office or general storage use.
Water / mechanical room	Ground, rear left	27 m2	Domestic booster / water equipment; fire water storage separate.
Main electrical room	Ground, rear right	27 m2	Exterior access; dry location; reserve working clearances.
Server / UPS	Ground, rear right	27 m2	Dedicated cooling and access; final UPS/battery separation to review.
BMS / security	Ground, inner right	25.2 m2	Operator desks and displays; separate from electrical switchgear.
AHU / ventilation rooms	First + second, rear right	27 m2 each	Maintenance and replacement routes; concept allowance only.
IDF / floor distribution	All floors, inner right	12.6 m2 each	Concept shared zone; separate IT/electrical areas as required.
Dry + HVAC risers	All floors, inner right	12.6 m2 each	Separate shaft compartments; not an occupied room.
Wet riser	All floors, inner left	4.8 m2 each	Stack plumbing; access panels from janitor area.
Toilets + pantry	Same positions, all floors	See plans	Keep wet services aligned and away from electrical/IT rooms.

Outside the model: reserve an accessible rear service yard for a fire-water tank (if required), heat rejection equipment / condensers, and possible standby generator. Their footprints are not yet sized or shown as fitted equipment.

Roof: no plant loads are placed on the glazed roof. Any roof-mounted HVAC needs an engineered support zone, access, drainage and maintenance space. A ground-level plant yard is the starting option.

Ceiling zones: reserve continuous service routes above corridors. Floor height, ceiling void depth, duct size and penetration positions must be coordinated before detailed HVAC and BMS modelling.

Build from this layout

A concept you can inspect now, then align with the original building file.

What is preserved

The two-wing arrangement, central atrium, three occupied levels and approximate central lift/stair location. The envelope is simplified; facade bays, balconies and columns are not reconstructed.

What is added

Room partitions, indicative furniture, a corridor network, stacked service zones and two rear stair-core reservations. Service rooms will need opaque, appropriately separated enclosures within the glazed shell.

Circulation intent

Main wing corridors are 2.4 m wide; secondary spines and atrium side galleries are 2.0 m in this concept. Rooms have door openings to corridors, galleries or outside. Door widths are placeholders. Shaft access is by service panels.

Important limits

The new rear stairs are space reservations, not designed escape stairs. The central stair is a location placeholder. Occupant load, exits, travel distances, stair discharge, accessibility and atrium smoke control remain unverified.

Files included

Three editable SVG plans, three plan PNGs, a PDF booklet, per-floor GLB cutaways, stacked and exploded GLB models, Room_Layout.json and the generation script.

How to use the 3D model

Import a GLB into Blender or a compatible 3D application. Units are metres. Floors are grouped; room IDs appear in object names. The model uses low cutaway walls for visibility and simple block furniture.

Next coordination step

Provide the original building model (.blend, .ifc, .glb, .fbx or other available format) and one confirmed dimension. Align the concept to actual slabs, structural columns, facade openings, stairs and lift before finalising partitions.

Next BMS step

Use the room IDs as digital-twin zones. Later add temperature, humidity, CO2, occupancy and energy points, then equipment/status links. Fire-system interfaces and controls require their own approved design.

Technical reference for the next stage: NFPA 20, Standard for the Installation of Stationary Pumps for Fire Protection (2025), official NFPA overview. It addresses pump selection and installation; no compliance assessment or hydraulic design is included here.

NFPA 20 official overview

A proposed wayfinding layout, pending egress verification

This drawing places signs at candidate exits, corridor decisions, gallery connections and selected larger-room doors. It does not establish that the building has adequate exits or that every shown route is safe in a fire.

BASIS USED

Confirmed by the user: NFPA 101 is the design basis; both rear stairs exist and are enclosed. The adopted edition, local amendments and approving authority are not yet identified.

Source limitations: the supplied concept assumes a 40 x 24 m envelope. Door sizes, stair geometry and columns are placeholders. Stair A / B labels in the base drawing are retained for traceability.

Route intent: each wing leads to its own rear stair. Upper floors descend to ground through those stairs. The main ground entrance is a candidate final exit. Atrium galleries connect to the wing corridors only if permitted by the approved fire strategy.

VERIFY BEFORE INSTALLATION

V1 / Final discharge: confirm the two rear stair doors and main entrance are usable egress doors. Survey continuous outdoor paths to a public way; no assembly point or safe exterior route can be fixed from this drawing.

V2 / Stairs: check fire resistance, continuity, door rating/closers, clear widths, landings, direction of descent, re-entry and discharge level.

V3 / Central core: the central stair is not counted as an exit. No evacuation route is assigned to the lift.

V4 / Atrium: check enclosure, sprinklers, smoke management and tenability. Even green wing routes may be exposed through their open connection to the atrium.

BUILDING-WIDE EGRESS CHECKS ARE STILL REQUIRED

Determine occupancy classification and occupant loads (including training/meeting uses), required number and remoteness of exits, travel distance, common path, dead ends, exit capacity and door hardware. Verify an accessible means-of-egress strategy and assistance/refuge provisions as applicable. Signage cannot correct a noncompliant route.

REFERENCE BASIS

NFPA 101, adopted edition to be confirmed: review Chapter 7 (means of egress, especially Section 7.10 marking), atrium provisions and the applicable occupancy chapter. The official edition page is below; the latest published edition is not automatically the locally adopted edition.

https://www.nfpa.org/codes-and-standards/all-codes-and-standards/101

Supporting explanation: Fire Protection Research Foundation, A Review of Dynamic Directional Exit Signage (2022), Section 2.1.1. Research guidance only; it does not replace the adopted code. The sign requirements and product selection must be checked by the project fire/life-safety designer.

Open the supporting NFPA-hosted research report

Match each sign to the direction occupants approach

Prefixes G-, 1- and 2- mean ground, first and second floor. Each row below applies to all three floors (42 mapped positions). Front/rear and left/right refer to the drawing orientation, not geographic directions.

SUFFIX	LOCATION	PROPOSED SIGN / MOUNT	DIRECTION AND VIEWING INTENT
G-20 / 1-17 / 2-17	Left front spine / main corridor	Directional EXIT at T-junction	Face occupants arriving from front rooms; turn toward left rear spine and Stair A.
G-21 / 1-18 / 2-18	Right front spine / main corridor	Directional EXIT at T-junction	Face occupants arriving from front rooms; turn toward right rear spine and Stair B.
01	Stair A entrance	EXIT over door, corridor side	Visible from both ends of left rear spine. Door is left of occupants arriving from main corridor.
02	Stair B entrance	EXIT over door, corridor side	Visible from both ends of right rear spine. Door is right of occupants arriving from main corridor.
03	Left front spine	Directional EXIT, high level	Read from front rooms moving toward main corridor; continue to corridor, then toward Stair A.
04	Right front spine	Directional EXIT, high level	Read from front rooms moving toward main corridor; continue to corridor, then toward Stair B.
05	Left main / rear spine turn	Directional EXIT at junction	For arrivals from inner wing: turn into rear spine toward Stair A; provide visible face before the turn.
06	Right main / rear spine turn	Directional EXIT at junction	For arrivals from inner wing: turn into rear spine toward Stair B; provide visible face before the turn.
07	Left rear spine	Directional EXIT repeater	Face approaches from main corridor and adjacent room doors; continue to Stair A entrance.
08	Right rear spine	Directional EXIT repeater	Face approaches from main corridor and adjacent room doors; continue to Stair B entrance.
09	Left inner wing / atrium junction	Directional EXIT, multiple approaches	Toward Stair A along left main corridor. Ground only: separate face toward main entrance if approved.
10	Right inner wing / atrium junction	Directional EXIT, multiple approaches	Toward Stair B along right main corridor. Ground only: separate face toward main entrance if approved.
11	Left rear atrium-side gallery	Directional EXIT at gallery bend	From rear gallery / rear room door: toward main corridor, then Stair A. Validate atrium exposure.
12	Right rear atrium-side gallery	Directional EXIT at gallery bend	From rear gallery / rear room door: toward main corridor, then Stair B. Validate atrium exposure.

Symbols mark locations, not installation dimensions. Over-door signs face the exit-access side. Corridor signs may need ceiling suspension, perpendicular mounting or separate faces. Set the printed chevron for the actual viewer; a plan-up travel arrow does not prescribe an upward arrow on the product.

Complete the route through the doors and down the stairs

ID	LOCATION	PROPOSED SIGN / DIRECTION
G-13	Main entrance	EXIT above the interior side of the exterior opening; confirm compliant door arrangement and discharge.
G-14 / G-15	Rear stair exterior doors	EXIT on stair side at ground discharge. Paths beyond these doors require a site plan (V1).
G-16 / G-17 / G-18	Fire pump / water-mechanical / electrical rooms	EXIT above each shown exterior door, visible from inside that room. Not a through-route for office occupants.
G-19	Training room G-05	EXIT above the room-side door to the right front spine; turn toward main corridor, then G-06 / Stair B.
1-13 / 1-14; 2-13 / 2-14	Front gallery ends, upper floors	Directional EXIT toward the corresponding side gallery, then the main corridor and rear stair.
1-15 / 1-16	Team workspaces A / B	EXIT above the room-side corridor door. Validate visibility from the farthest desks.
2-15 / 2-16	CEO office / boardroom	EXIT above the room-side corridor door. Check furniture and partitions do not obscure the sign.

57 MAPPED SIGN POSITIONS

Ground: 21. First: 18. Second: 18. This is a preliminary coordination count, not an order quantity. Extra sign faces, intermediate repeaters, stair-interior signs and accessibility signs are not quantified.

INSTALLATION & WALK-THROUGH CHECK

Visibility: walk from every occupied room, toilet, pantry and service area. At every exit-access decision, the next sign must be visible and its direction clear. Check both approaches to turns and door-open/door-closed conditions. Add repeaters where walls, furniture or rated viewing distances require them.

Inside stairs: provide descent guidance and discharge identification as needed; identify stair and floor levels as required by the adopted code. Prevent occupants continuing below the discharge level where a basement exists. Detailed landings are absent, so these signs are not located here.

Sign specification: select listed illuminated EXIT units (e.g. UL 924 as applicable), with compliant chevrons and emergency operation; verify the adopted NFPA 101 requirements, including the applicable 90-minute emergency supply. Coordinate mounting, language, color and accessibility signage with the authority.

Related systems: design emergency route lighting separately. Verify locking, free egress and fire-door function. Mark misleading non-exit doors by their actual use or approved NOT AN EXIT wording where required. Test signs and emergency power; maintain records.

RELEASE GATE: surveyed plan + approved fire strategy + verified final discharge + signed-off visibility review.